$$(\rho \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} f)(x) = z^x f(x)$$

(note that z^x makes sense since $z^p = 1$).

(a) Show that such a representation exists and is unique, and compute $\rho(g)$ for all $g \in G$.

(b) Denote this representation by R_z. Show that R_z is irreducible if and only if $z \neq 1$.

(c) Classify all 1-dimensional representations of G. Show that R_1 decomposes into a direct sum of 1-dimensional representations, where each of them occurs exactly once.

(d) Use (a)—(c) and the "sum of squares" formula to classify all irreducible representations of G.

Problem 4.12.3. Let V be a finite dimensional complex vector space, and let $GL(V)$ be the group of invertible linear transformations of V. Then S^nV and Λ^mV ($m \leq \dim(V)$) are representations of $GL(V)$ in a natural way. Show that they are irreducible representations.

Hint: Choose a basis $\{e_i\}$ in V. Find a diagonal element H of $GL(V)$ such that $\rho(H)$ has distinct eigenvalues (where ρ is one of the above representations). This shows that if W is a subrepresentation, then it is spanned by a subset S of a basis of eigenvectors of $\rho(H)$. Use the invariance of W under the operators $\rho(1 + E_{ij})$ (where E_{ij} is defined by $E_{ij}e_k = \delta_{jk}e_i$) for all $i \neq j$ to show that if the subset S is nonempty, it is necessarily the entire basis.

Problem 4.12.4. Recall that the adjacency matrix of a graph Γ (without multiple edges) is the matrix in which the ijth entry is 1 if the vertices i and j are connected with an edge, and zero otherwise. Let Γ be a finite graph whose automorphism group is nonabelian. Show that the adjacency matrix of Γ must have repeated eigenvalues.

Problem 4.12.5. Let I be the set of vertices of a regular icosahedron ($|I| = 12$). Let $F(I)$ be the space of complex functions on I. Recall that the group $G = A_5$ of even permutations of five items acts on the icosahedron, so we have a 12-dimensional representation of G on $F(I)$.